

Python & Akustik: Ambisonics Player

Ambisonics has established itself as a widely used format for spatial audio, primarily due to its flexibility and compatibility with head-related transfer functions (HRTFs), making it a popular means of binaural reproduction via headphones. Two established approaches to binaural decoding are rendering using interposed virtual loudspeakers [1] and direct processing in the spherical harmonics domain [2].

The goal of this project is to develop an Ambisonics player for conducting binaural hearing tests in Python. The package should read Ambisonics files (ANC/SN3D) and make them binaurally playable using one of the two approaches mentioned above. The following core functions are required:

- **File selection** via source path and detection of the Ambisonics order
- **Playback** of the file with selectable Ambisonics playback order
- **Binaural decoding** using a predefined HRTF dataset
- Basic **transport controls** (play, pause, stop, volume adjustment, loop function, and start offset in samples)
- **GUI** with file selection and transport control elements

Ideally, the software could be extended to include dynamic head tracking, enabling real-time rotation of the Ambisonics scene. Existing interfaces such as PyHeadTracker or similar solutions could be used to connect hardware trackers. Furthermore, it would be desirable to be able to load HRTF datasets in SOFA format, thus supporting the use of various HRTFs.

The software should be designed according to the principles of object-oriented programming so that it can be seamlessly integrated as a standalone module into larger test environments or other projects. The code should be managed in a Git repository and fully documented.

Additional References:

Pyshroom (SH Decoding) (<https://pypi.org/project/pyshroom/>)

Mathworks (Virtual Loudspeaker Approach)

(<https://www.mathworks.com/help/audio/ug/ambisonic-binaural-decoding.html>)

IEM Plugin-Suite (<https://plugins.iem.at/docs/pluginDescriptions/#multiencoder>)

spharpy (<https://spharpy.readthedocs.io/en/stable/spharpy.html>)

[1] Schörkhuber, C., Zaunschirm, M., & Höldrich, R. (2018). "Binaural Rendering of Ambisonic Signals via Magnitude Least Squares." Proceedings of the DAGA, Munich, pp. 339–342.

[2] Zotter, F. & Frank, M. (2012). "All-Round Ambisonic Panning and Decoding." Journal of the Audio Engineering Society, 60(10), 807–820.